



Write the Steps

Quick facilitation notes & answer key

Goal

Students write an algorithm: they order a step bank into the steps for brushing teeth, explain why that order is needed, then write their own bedtime algorithm. Writing the ordered steps of a task is creating a sequential algorithm (C3.1); explaining why the order matters is C3.2.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: cut the step bank apart so students can arrange the cards before copying them onto the lines.

Run it (about 20 min)

1. I DO: read the step bank. 'I write the FIRST step on line 1 — wet the brush — then keep going.'
2. WE DO: Exercise 1 — write wet the brush, put on toothpaste, brush your teeth, rinse your mouth (B, D, A, C). Exercise 2 — first is 'wet the brush'; toothpaste must come before brushing.
3. YOU DO: Exercise 3 — students write their own 3-4 step bedtime algorithm in order.
4. Connect: 'Writing the steps in order is writing code — coders plan the order so the job works.'

Answer key (modelled by the run-gate)

Ex 1 — Brush your teeth in order: wet the brush (B), put on toothpaste (D), brush your teeth (A), rinse your mouth (C).

Ex 2 — 'Wet the brush' must be first; you cannot brush before the toothpaste is on, so put on toothpaste comes before brush your teeth. Swapping them breaks the algorithm.

Ex 3 — open. A sample bedtime algorithm: put on pyjamas, brush your teeth, read a book, turn off the light. Accept any 3-4 steps in a sensible order (an action that depends on another comes after it).

Watch for & differentiate

Common slip in Ex 3: steps with no clear order — ask 'which one has to happen before another?' to surface dependencies.

Simplify: give a starter step bank for the bedtime task instead of a blank write.

Extend: ask students to trade algorithms with a partner and follow each other's steps to test them.

Success indicator: the child writes B, D, A, C for brushing and a sensible, correctly-ordered bedtime algorithm.